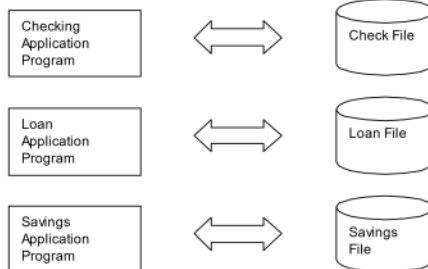


REVIEW OF DATABASES

ITFN 2214 – WEB APPLICATION DEVELOPMENT

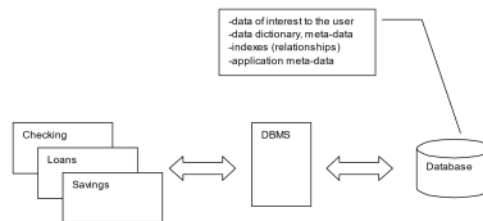
FILE PROCESSING

- Data vs. Information
- Information is critical
- Disadvantages
 - Data separate and isolated
 - Data duplication
 - Applications dependent on file formats
 - Complex objects hard to represent



DATABASE PROCESSING

- Intermediary between apps and data
- Allows integration of data
- Reduces redundancy
- Application independence
- More vulnerable



WHAT IS A DATABASE?

- A database is a **self-describing** collection of integrated (**related**) records.
- Self-describing
 - The data dictionary contains a complete description of the database structure
- Collection of Integrated Records
 - Relationships among related records must be stored in the data dictionary or the data itself
- What is a DBMS?
 - Set of programs used to define, process, and administer the database and its applications

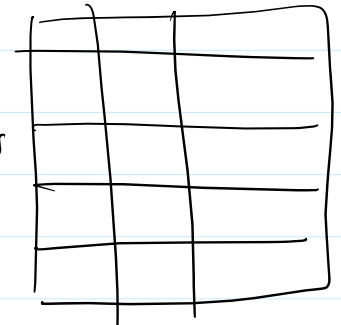
RELATIONAL DBMS

- Developed and published by E.F. Codd in 1970
- A two-dimensional table with rows and columns with characteristics of:
 - Each row is unique
 - Column names are unique
 - Cells in the table must be single-valued
 - No order on the rows
 - No order on the columns

TERMS YOU SHOULD KNOW

- Attribute - characteristics of an entity (i.e., column)
- Tuple = row
- Relation - table
- Candidate key - the minimal set of columns or group of columns that uniquely identify a row
- Primary key - chosen candidate key
- Foreign key
- Referential integrity
- SQL

Superkey - all columns or groups of columns that uniquely identify a row of a table



ADVISER_LIST

AdviserID	AdviserLastName	AdviserFirstName	Department	Phone	Office	StudentNumber	StudentLastName	StudentFirstName
E0062	Baker	Linda	Accounting	301-557-1011	Higgins 305	S0023	Andrews	Matthew
E0062	Baker	Linda	Accounting	301-557-1011	Higgins 305	S0055	Barnes	Ilse

AdviserID	AdviserLastName	AdviserFirstName	Department	Phone	Office	StudentNumber	StudentLastName	StudentFirstName
E0062	Baker	Linda	Accounting	301-557-1011	Higgins 305	S0023	Andrews	Matthew
E0062	Baker	Linda	Accounting	301-557-1011	Higgins 305	S0065	Bousaid	Ibrahim
E0234	Taing	Susan	Accounting	301-557-1011	Higgins 315	S0083	Jiang	Terry
E0345	Tran	Ken	InfoSystems	301-557-1041	Codd 412	S0132	Thompson	James
E0017	Valdez	Esteban	Chemistry	301-557-1031	Valance 251	S0154	Reyes	Elisa
E0084	Yeats	Bill	InfoSystems	301-557-1041	Codd 423	S0167	Lacroix	Gabriel

relation

YOUR TURN

- Draw a picture of a table. Put a few rows of data into the table.
- Use your table to define these terms:
 - Attribute
 - Tuple
 - Relation
 - Candidate key
 - Primary key

ADVISER → PK
 AdviserID | Last
 E0062 | Baker . . .

YOUR TURN (2)

- Draw a picture of two tables. Put a few rows of data in each.
- Use your picture to define these terms
 - Primary key
 - Foreign key
 - Referential integrity

Functional Dependencies

determination : the state in which knowing the value of an attribute/column makes it possible to determine the value of another.

$$\text{CookieCost} = \text{numBoxes} * 5;$$

functional dependency is a relationship between columns in which the value of one or more columns determines the value of one or more other columns.

← **determinant**
 $A \rightarrow B \leftarrow$ **dependent**
"A determines B"

$$\text{numBoxes} \rightarrow \text{cookieCost}$$

$$\underline{\text{Quantity} * \text{Unit Price} = \text{Final Price}}$$

Quantity * Unit Price = Final Price
 (Quantity, Unit Price) → Final Price

If I tell you one specific fact, can you respond w/ a unique associated fact?

Object Color	Weight	Shape
Red	5	Ball
Blue	5	Cube
Yellow	7	Cube ✓

Object Color → Weight
 Object Color → Shape

Object Color → Weight, Shape

Another e.g.

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E0084	Yeats	Bill	InfoSystems	301-557-1041	Codd 423	S0167	Lacroix	Gabriel

AdviserID → AdviserLastName, AdviserFirstName, Dept, Phone, Office
 StudentNumber → StudentLastName, StudentFirstName, AdviserID

Another e.g. → Assume PetNames, ^{Owner Phone, Owner Email} are unique & PetDOB is not

	A	B	C	D	E	F	G	H
	PetName	PetType	PetBreed	PetDOB	OwnerLastName	OwnerFirstName	OwnerPhone	OwnerEmail
1	King	Dog	Std. Poodle	27-Feb-16	Downs	Marsha	201-823-5467	Marsha.Downs@somewhere.com
3	Teddy	Cat	Cashmier	1-Feb-13	James	Richard	201-735-9812	Richard.James@somewhere.com
4	Fido	Dog	Std. Poodle	17-Jul-17	Downs	Marsha	201-823-5467	Marsha.Downs@somewhere.com
5	AJ	Dog	Collie Mix	5-May-17	Frier	Liz	201-823-6578	Liz.Frier@somewhere.com
6	Cedro	Cat	Unknown	6-Jun-14	James	Richard	201-735-9812	Richard.James@somewhere.com
7	Woolley	Cat	Unknown	???	James	Richard	201-735-9812	Richard.James@somewhere.com
8	Buster	Dog	Border Collie	11-Dec-13	Trent	Miles	201-634-7865	Miles.Trent@somewhere.com
9	Jiddah	Cat	Abyssinian	1-Jul-10	Evans	Hilary	201-634-2345	Hilary.Evans@somewhere.com

PetName → PetType, PetBreed, PetDOB, Owner Last, First, Phone, Email

~~Owner LastName~~ →

Phone → Email, First, Last

Email → Phone, First, Last

Phone → Email, First, Last
Email → Phone, First, Last

Student Number	LastName	First Name	DormName	Dorm Cost
100	Smith	Terry	Stephens	3500
200	Johnson	Jeff	Alexander	3800
300	Albernathy	Susan	Horan	4000
400	Smith	Susan	Alexander	3800
500	Wilcox	John	Stephens	3500
600	Webber	Carl	Horan	4000
700	Simon	Carol	Stephens	3500

Student Number → LastName, LastName, DormName, Dorm Cost
Dorm Name → Dorm Cost